

Cross-Layer Fingerprint Impersonation Detector

Project Features & Function List

This document lists the proposed functions for the semester project, including the instructor-requested accuracy measurement, audit logging, CSV dataset generation, and reporting features. The core design follows the project's existing Capture → Extract → Compare → Explain workflow.

1. Traffic Capture Functions

Function	Purpose
<code>capture_live_traffic()</code>	Captures network traffic from a live network interface.
<code>load_pcap()</code>	Loads previously captured network traffic from a PCAP file.
<code>create_flows()</code>	Groups packets that belong to the same network connection/flow.

2. Fingerprint Extraction Functions

Function	Purpose
<code>extract_ja4(flow)</code>	Extracts the JA4 TLS fingerprint.
<code>extract_ja4t(flow)</code>	Extracts the JA4T TCP/OS fingerprint.
<code>extract_ja4h(flow)</code>	Extracts the JA4H HTTP fingerprint.
<code>extract_all_fingerprints(flow)</code>	Collects JA4, JA4T, and JA4H for one flow.

3. Identity & Reference Functions

Function	Purpose
<code>load_reference_table()</code>	Loads the reference information used to interpret fingerprints.
<code>lookup_tls_identity(ja4)</code>	Determines the client/OS identity suggested by the TLS fingerprint.
<code>lookup_tcp_identity(ja4t)</code>	Determines the OS/network-stack identity suggested by TCP behavior.
<code>lookup_http_identity(ja4h)</code>	Determines the client/browser identity suggested by HTTP behavior.

4. Cross-Layer Detection Functions

Function	Purpose
<code>compare_fingerprints()</code>	Compares the identities suggested by TLS, TCP, and HTTP.
<code>detect_mismatch()</code>	Identifies which layers disagree and what type of mismatch occurred.
<code>generate_verdict()</code>	Labels a flow as CONSISTENT or FLAGGED.

5. Explanation Functions

Function	Purpose
<code>generate_explanation()</code>	Creates a readable explanation of the detected mismatch.
<code>explain_flag_with_llm()</code>	Uses an LLM to turn a technical flag into a short plain-English explanation.

6. Audit Logging Functions — NEW

Function	Purpose
<code>create_audit_record()</code>	Creates a complete record for every analyzed network flow.
<code>write_audit_log()</code>	Writes analysis results to an <code>audit_log.csv</code> file.
<code>load_audit_log()</code>	Loads previously recorded audit entries.
<code>export_audit_log()</code>	Exports analysis results as CSV for review or further processing.
<code>search_audit_logs()</code>	Filters audit records, such as showing only flagged connections.

7. Accuracy & Evaluation Functions — NEW

Function	Purpose
<code>assign_actual_label()</code>	Stores the ground-truth label, such as NORMAL or SPOOFED, for evaluation data.
<code>calculate_accuracy()</code>	Calculates the percentage of correct predictions.
<code>calculate_precision()</code>	Measures how many flagged connections were actually spoofed.
<code>calculate_recall()</code>	Measures how many actual spoofed connections were detected.
<code>calculate_f1_score()</code>	Combines precision and recall into one evaluation score.
<code>generate_confusion_matrix()</code>	Generates TP, TN, FP, and FN results for detailed evaluation.

8. Dataset Functions — NEW

Function	Purpose
<code>build_dataset_from_logs()</code>	Converts audit logs into a structured dataset.
<code>validate_dataset()</code>	Checks for missing values, duplicates, invalid labels, and incomplete records.
<code>split_dataset()</code>	Separates data into training/testing sets if future machine-learning experiments are performed.
<code>export_dataset()</code>	Exports a clean fingerprint dataset as CSV.

9. Reporting Functions — NEW

Function	Purpose
<code>generate_report()</code>	Creates an overall summary of detection and evaluation results.
<code>generate_flow_report()</code>	Shows detailed results for individual network flows.
<code>generate_statistics()</code>	Calculates overall statistics such as total flows, flagged flows, and consistent flows.

Recommended CSV Audit Log Fields

Each analyzed flow should produce one row. The same CSV can act as both an audit trail and a source dataset for evaluation.

Field	Purpose
timestamp	Time the flow was analyzed
flow_id	Unique identifier for the network flow
source_ip / destination_ip	Network endpoints
source_port / destination_port	Connection ports
protocol	Network protocol
ja4	TLS fingerprint
ja4t	TCP/OS fingerprint
ja4h	HTTP fingerprint
tls_identity	Identity suggested by TLS
tcp_identity	Identity suggested by TCP
http_identity	Identity suggested by HTTP
verdict	CONSISTENT or FLAGGED
mismatch_layer	Layer responsible for the mismatch
explanation	Plain-English reason for the result
actual_label	Known ground-truth label for evaluation
correct	Whether the prediction was correct

Evaluation Metrics

Accuracy = $\text{Correct Predictions} \div \text{Total Predictions} \times 100$

Precision = Of the connections flagged as spoofed, how many were actually spoofed.

Recall = Of all actual spoofed connections, how many were detected.

F1-score = Combined measure of precision and recall.

Overall System Flow

Network Traffic → Capture → Flow Creation → JA4/JA4T/JA4H Extraction → Identity Lookup → Cross-Layer Comparison → Consistent/Flagged → Explanation → Audit CSV → Dataset & Accuracy Evaluation → Report